

EDUCATION

- Sept. 2023 - **Ph.D. student in Computer Science, Rutgers University**
Present Advised by Prof. Xiong (Leo) Fan, Department of Computer Science.
- Sept. 2019 - **M.S. in Computer Science, Rutgers University, GPA:3.85**
May. 2023
- Sept. 2012 - **B.E. in Software Engineering, Sun Yat-sen University, China**
Jun. 2016

PUBLICATIONS

- [IEEE Euro S&P 2026] **Yuange Li**, and Xiong Fan, SUMMER: Recursive Zero-Knowledge Proofs for Scalable RNN Training, In: *IEEE European Symposium on Security and Privacy, 2026, Accepted*

RESEARCH PROJECTS

Verifiable Information Retrieval System, In Progress

- Developing cryptographic verification techniques for correctness and integrity of queries over large-scale, dynamically evolving data systems.
- Designing scalable architectures for verifiable data systems that support incremental updates and efficient consistency guarantees.
- Optimizing proof-generation pipelines to maintain succinctness and low latency under high-throughput workloads.

SUMMER: Recursive Zero-Knowledge Proofs for Scalable RNN Training, [ePrint 2025/1688]

- Designed and implemented the first end-to-end Zero-Knowledge Proof of Training (zkPoT) system for Recurrent Neural Networks (RNNs).
- Developed a unified verification pipeline for Backpropagation Through Time (BPTT), combining sumcheck-based proofs for linear layers and LogUp-style lookup arguments for non-linear activations.
- Achieved constant proof size and verification time independent of training iterations via recursive proof composition (IVC), outperforming state-of-the-art baselines by 8.5x in proving speed and 11.6x in memory efficiency.

Ontology-Based Decision Making for Autonomous Vehicles, Visiting Scholar @ Technical University of Munich

- Leveraged Web Ontology Language (OWL) to design semantic reasoning models for Advanced Driver Assistance Systems (ADAS), enhancing environmental perception capabilities.
- Engineered hierarchical traffic situation ontologies to support complex autonomous navigation and conflict resolution at dynamic road intersections.
- Formulated an ontology integration framework to improve high-level decision-making processes for autonomous systems in unstructured traffic environments.

WORK EXPERIENCE

- Jun. 2025 - **Research Engineer Intern, Cysic, Remote**
Present
 - Developed **SUMMER**, an end-to-end Zero-Knowledge Proof of Training (zkPoT) system for RNNs that achieves 8.5x performance improvement over baselines.
 - Exploring and prototyping next-generation zero-knowledge primitives for Large Language Models (LLMs) and other advanced cryptographic methods.
- Jun. 2024 - **Research Engineer Intern, Cysic, Remote**
Aug. 2024
 - Conducted research and development focused on Halo2 and Plonky2 for cryptographic primitives in decentralized applications.
 - Designed and optimized zk-SNARK protocols to improve proof generation and verification efficiency in zero-knowledge proof systems.
 - Tested, analyzed, and applied open-source libraries, including SP1, RISC ZERO, Intmax2, EZKL, and Expander for zero-knowledge solutions.

- Aug. 2020 - **Senior Solutions Manager, SenseTime, Shenzhen, China**
- Aug. 2022
- Designed tailored solutions and processed transformation recommendations to meet current and future business.
 - Assessed customer strategic needs and discover key business requirements to propose company's products, solutions, and services, articulating advantages and compelling business value.
 - Collaborated with various product teams locally and overseas to understand the products as well as to propose the best suit-client-need solution.
 - Provided timely, prioritized, and complete customer-based feedback to Product Management, Research, Sales, Engineering and/or Support regarding client's business cases, requirements and issues.
- Aug. 2016 - **Android Development Engineer, Huawei Technologies CO., LTD, Shenzhen, China**
- Jul. 2019
- Developed core modules for Huawei AppGallery, achieving over one trillion app interactions.
 - Implemented Android Client of Huawei AppGallery Essential Apps, used by more than 160 million users monthly, ensuring robust performance and reliability.
 - Built and maintained multi-language support for applications deployed in 163 countries.
 - Collaborated with cross-functional teams to optimize the performance and deployment of distributed applications.
 - Integrated backend APIs with global cloud services to ensure scalability and secure data management.
- Apr. 2017 - **VR Development Engineer, Huawei Technologies CO., LTD, Shenzhen, China**
- Dec. 2017
- Led a team of seven in designing and implementing Huawei VRStore Client, integrating 3D user interface components.
 - Enhanced system performance through optimization of VR data streaming on cloud infrastructures, improving data throughput and latency.
 - Conducted extensive testing and validation for cross-platform compatibility, ensuring seamless user experience.
- Jul. 2015 - **Software Engineer Intern, Huawei Technologies CO., LTD, Shenzhen, China**
- Spet. 2015
- Maintained and optimized the Huawei AppGallery Android client, debugging and deploying updates to enhance performance and stability.
 - Collaborated with senior engineers to implement improvements for multi-region deployment, enhancing application reach and usability.

AWARDS & CERTIFICATES

2012-2013 **University Fellowship, Sun Yat-sen University**

TEACHING EXPERIENCE

- Spring 2026 **Teaching Assistant, CS442: Introduction to Cryptography, Rutgers University**
- Fall 2025 **Teaching Assistant, CS344: Design and Analysis of Computer Algorithms, Rutgers University**
- Fall 2024 **Teaching Assistant, CS512: Introduction to Data Structures and Algorithms, Rutgers University**
- Spring 2024 **Teaching Assistant, CS419: Computer Security, Rutgers University**
- Fall 2023 **Teaching Assistant, CS206: Introduction to Discrete Structures II, Rutgers University**
- Spring 2015 **Teaching Assistant, Communication Theory, Sun Yat-sen University**

ACADEMIC SERVICE

External
Reviewer

FC'24, Crypto'25, IEEE S&P'26, Crypto'26

PROGRAMMING LANGUAGES

Java, Python, C++, C, Rust